QUASI-COMPACTNESS CHECKED ON AFFINE OPENS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3812-3840, following the statement at 3800-3810.

Editorial note. The bridging steps 'a finite union of affine opens is quasi-compact' and '(2) implies (3)' are standard/trivial and left implicit by the author; the equivalence chain runs (3) -> finite-union -> (1) -> finite-union -> (2).

Allowed standard prerequisites (editorial).

- Standard opens of Spec are functorial. Algebra, Lemma algebra-lemma-spec-functorial (cross-chapter, cited at ~3823). Preimage of a standard open under a map of affines is standard, hence affine.

Source reference. Original reference 00E2: focused-seeds/geometry/S9-schemes-quasi-compact-affine/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. The following are equivalent*

(1) *$f : X \to S$ is quasi-compact,*
(2) *the inverse image of every affine open is quasi-compact, and*
(3) *there exists some affine open covering $S = \bigcup_{i \in I} U_i$ such that $f^{-1}(U_i)$ is quasi-compact for all i.*

Proof. Suppose we are given a covering $S = \bigcup_{i \in I} U_i$ as in (3). First, let $U \subset S$ be any affine open. For any $u \in U$ we can find an index $i(u) \in I$ such that $u \in U_{i(u)}$. As standard opens form a basis for the topology on $U_{i(u)}$ we can find $W_u \subset U \cap U_{i(u)}$ which is standard open in $U_{i(u)}$. By compactness we can find finitely many points $u_1, \ldots, u_n \in U$ such that $U = \bigcup_{j=1}^{n} W_{u_j}$. For each j write $f^{-1}U_{i(u_j)} = \bigcup_{k \in K_j} V_{jk}$ as a finite union of affine opens. Since $W_{u_j} \subset U_{i(u_j)}$ is a standard open we see that $f^{-1}(W_{u_j}) \cap V_{jk}$ is a standard open of V_{jk}, see Algebra, Lemma 00E2. Hence $f^{-1}(W_{u_j}) \cap V_{jk}$ is affine, and so $f^{-1}(W_{u_j})$ is a finite union of affines. This proves that the inverse image of any affine open is a finite union of affine opens.

Next, assume that the inverse image of every affine open is a finite union of affine opens. Let $K \subset S$ be any quasi-compact open. Since S has a basis of the topology consisting of affine opens we see that K is a finite union of affine opens. Hence the inverse image of K is a finite union of affine opens. Hence f is quasi-compact.

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Finally, assume that f is quasi-compact. In this case the argument of the previous paragraph shows that the inverse image of any affine is a finite union of affine opens. $\square$